

Packet Switching

Unit-III

Lecture -2

August 9, 2018

Session Objectives

Packet Switching

Unit-III

NETWORK- LAYER PERFOR- MANCE

IPV4 ADDRESSES

- To discuss about packet switching
- To study the datagram approach and the virtual-circuit approach

Session Outcomes

Packet Switching

Unit-III

NETWORK- LAYER PERFOR- MANCE

IPV4 ADDRESSES

At the end of this session, participants will be able to

- Discuss packet switching

Agenda

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

1 NETWORK-LAYER PERFORMANCE

2 IPV4 ADDRESSES

Packet Switching

Packet Switching

Unit-III

NETWORK- LAYER PERFOR- MANCE

IPV4 ADDRESSES

- Packet-switched network can use two different approaches to route the packets: the datagram approach and the virtual circuit approach

Datagram Approach: Connectionless Service

Packet Switching

Unit-III

NETWORK- LAYER PERFOR- MANCE

IPV4 ADDRESSES

- The packets in a message may or may not travel the same path to their destination.
- Each packet travelling an independent entity; there is no relationship between packets belonging to the same message.
- The switches in this type of network are called routers.
- A packet belonging to a message may be followed by a packet belonging to the same message or to a different message from the same or from a different source.
- Packet is routed based on the information contained in its header

Connectionless packet-switching

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

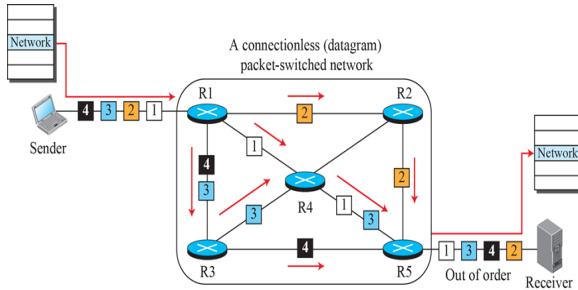


Figure: A connectionless packet-switched network

- Routes the packet based only on the destination address.
- Source address may be used to send an error message to the source if the packet is discarded

Forwarding process

Packet Switching

Unit-III

NETWORK- LAYER PERFOR- MANCE

IPV4 ADDRESSES

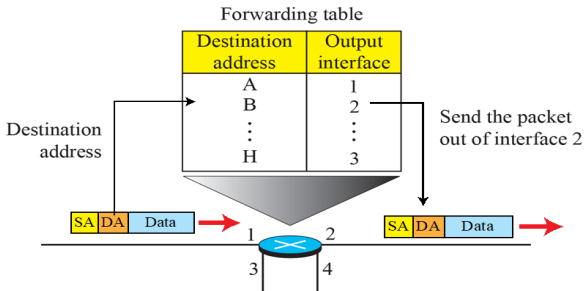


Figure: Forwarding process in a router when used in a connectionless network

Virtual-Circuit Approach: Connection-Oriented Service

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

- There is a relationship between all packets belonging to a message
- After connection setup, the datagrams can all follow the same path
- Packet has the source and destination addresses, also contain a **flow label**, a **virtual circuit identifier** that defines the virtual path the packet should follow

Virtual-circuit

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

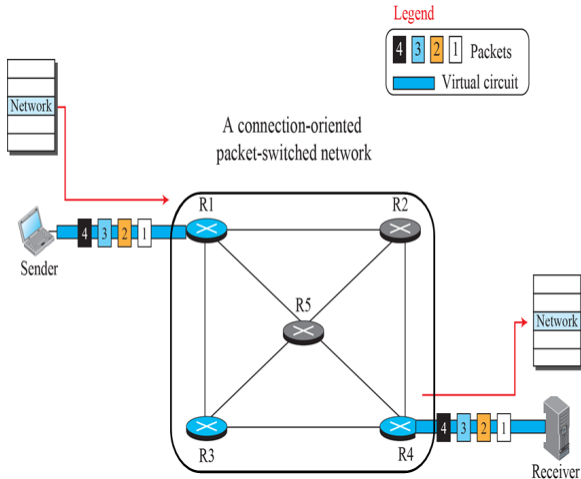


Figure: A virtual-circuit packet-switched network

Forwarding process

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

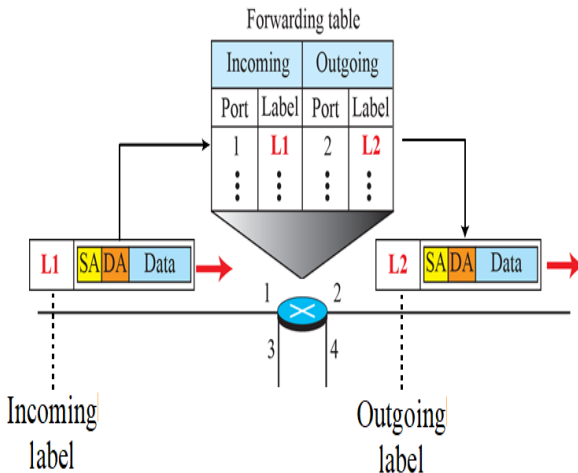


Figure: Forwarding process in a router when used in a virtual-circuit network

Setup Phase

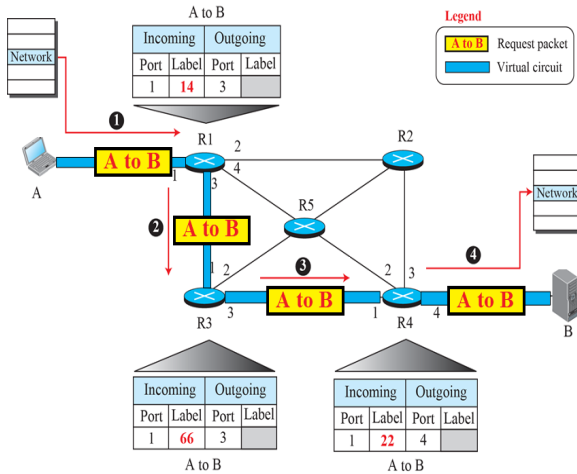
Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

- A router creates an entry for a virtual circuit
- Two auxiliary packets exchanged between the sender and the receiver: the request packet and the acknowledgement



Request packet

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

- 1 Source A sends a request packet to router R1.
- 2 Router R1 receives the request packet
 - knows that a packet from A to B **goes out through port 3**
 - **creates an entry** in its table for this virtual circuit,
 - **fill three of the four columns**
 - assigns the incoming port (1)
 - chooses an available incoming label (14)
 - the outgoing port (3)
 - does not yet know the outgoing label, done during the acknowledgement step
 - forwards the packet through port 3 to router R3.

Request packet

Packet Switching

Unit-III

NETWORK- LAYER PERFOR- MANCE

IPV4 ADDRESSES

- 1 Router R3 receives the setup request packet. The same events happen here as at router R1;
- 2 Router R4 receives the setup request packet. Again, three columns are completed: incoming port (1), incoming label (22), and outgoing port (4).
- 3 Destination B receives the setup packet,
— if it is ready to receive packets from A, it assigns a label to the incoming packets that come from A, in this case 77
— This label lets the destination know that the packets come from A, and not from other sources.

Acknowledgment Packet

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

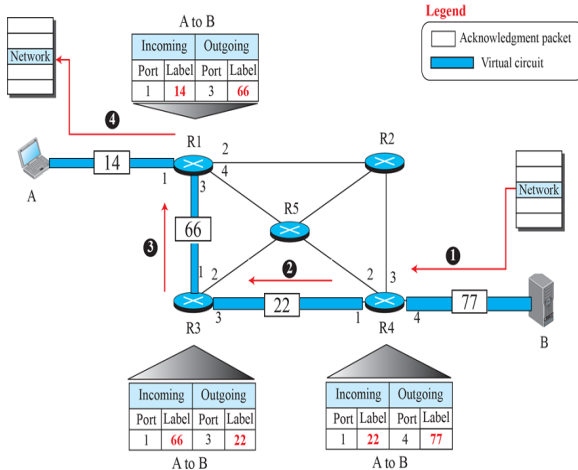


Figure: Sending acknowledgments in a virtual-circuit network

Data-Transfer Phase

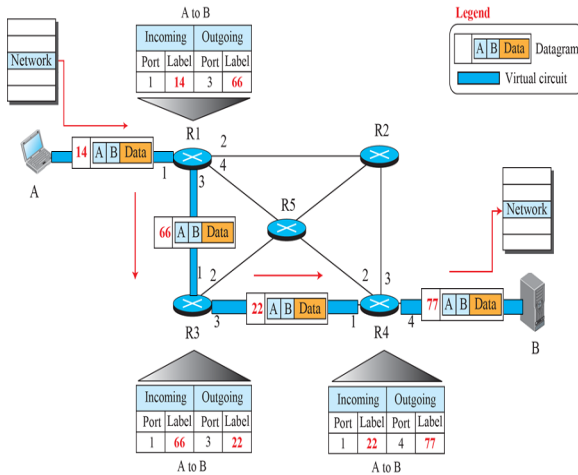
Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPv4 ADDRESSES

- After all routers have created their forwarding table for a specific virtual circuit, the network-layer packets belonging to one message can be sent one after another.



Teardown Phase

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

- source A, after sending all packets to B, sends a special packet called a teardown packet.
- Destination B responds with a confirmation packet. All routers delete the corresponding entries from their tables.

Presentation Outline

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

1 NETWORK-LAYER PERFORMANCE

2 IPV4 ADDRESSES

NETWORK-LAYER PERFORMANCE

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

Performance is measured in terms of delay, throughput, and packet loss.

- Transmission Delay:

$$Delay_{tr} = (\text{Packet length}) / (\text{Transmission rate}).$$

- Propagation Delay:

$$Delay_{pg} = (\text{Distance}) / (\text{Propagation speed}).$$

- Processing Delay:

$Delay_{pr}$ = **Time required to process a packet in a router or a destination host**

- Queuing Delay:

$Delay_{qu}$ = **The time a packet waits in input and output queues in a router**

Total delay =

$$(n + 1)(Delay_{tr} + Delay_{pg} + Delay_{pr}) + (n)(Delay_{qu})$$

Throughput

- The number of bits passing through the point in a second, which is actually the transmission rate of data at that point
- **Throughput = minimum ($TR_1, TR_2, \dots TR_n$)**

TR: Transmission rate



a. A path through three links

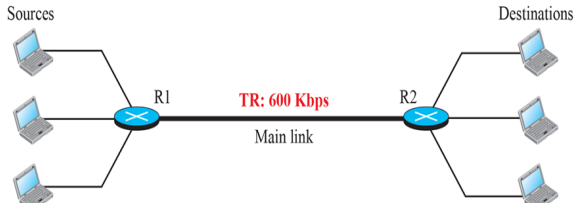


Throughput

- Internet backbone has a very high transmission rate, in the range of gigabits per second.
- The throughput is as minimum transmission rate of the two access links that connect the S and D.
- Router may collect the flow from several sources or distribute the flow between several sources

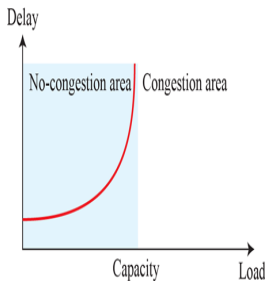


TR: Transmission rate

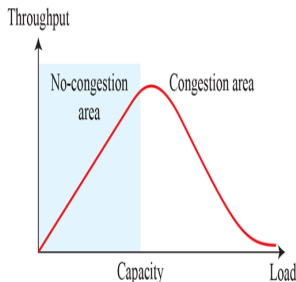


Congestion Control

- Congestion at the network layer is related to throughput and delay can be measured as functions of load.
- when the load reaches the network capacity, the delay increases sharply — becomes infinite when the load $\geq$ capacity
- Throughput declines sharply —the discarding of packets by the routers.



a. Delay as a function of load



b. Throughput as a function of load

Congestion control mechanisms

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

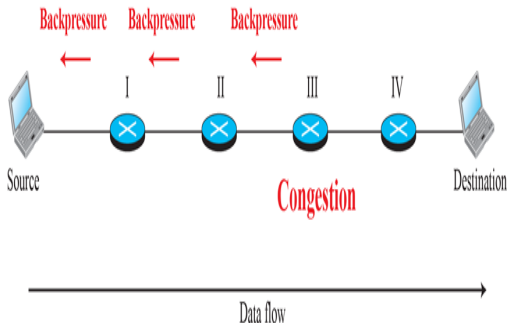
IPV4
ADDRESSES

Open-loop congestion control (prevention) :

- **Retransmission Policy:** may increase congestion —retransmission timers must be designed to optimize efficiency and at the same time prevent congestion.
- **Window Policy:** Go-Back-N window, when the timer for a packet times out, several packets may be resent. The Selective Repeat window, sends the specific packets that have been lost or corrupted
- **Acknowledgement Policy:** Sending fewer acknowledgements means imposing less load
- **Discarding Policy:** the quality of msg is still preserved and congestion is prevented or alleviated.
- **Admission Policy:** Switches in a flow first check the resource requirement of a flow before admitting it to the network.

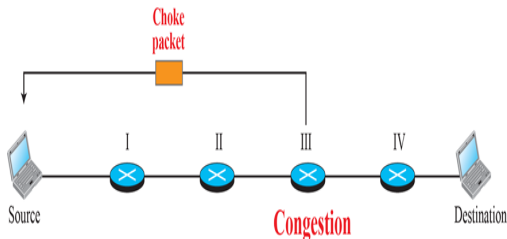
Closed-Loop Congestion Control

- **Backpressure:** refers to a congestion control mechanism in which a congested node stops receiving data from the immediate upstream node or nodes
- A node to node congestion control that starts with a node and propagates in the opposite direction towards source
- Can only be implemented in **virtual-circuit** — **cannot** be implemented in a **datagram** network



Closed-Loop Congestion Control: Choke Packet

- A packet sent by a node to the source to inform it of congestion.
- Backpressure: the warning is from one node to its upstream node;choke-packet method, the warning is from the router, which has encountered congestion, directly to the source station
- Intermediate nodes through which the packet has traveled are not warned



Closed-Loop Congestion Control

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

- **Implicit Signaling:** there is no communication between the congested node or nodes and the source. The source guesses that there is congestion somewhere in the network from other symptoms.
- **Explicit Signaling:**
 - Choke-packet method, a separate packet is used for this purpose; in the explicit-signaling method, the **signal is included in the packets that carry data.**
 - Explicit signaling can occur in either the forward or the backward direction

Presentation Outline

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

1 NETWORK-LAYER PERFORMANCE

2 IPV4 ADDRESSES

IPV4 ADDRESSES

Packet
Switching

Unit-III

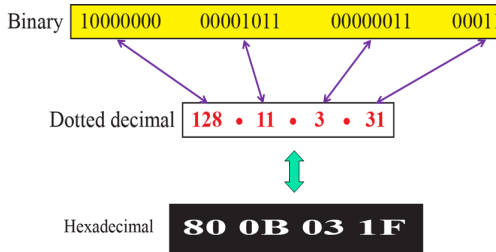
NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

- **Internet address or IP address:** Identifier used in the IP layer of the TCP/IP protocol suite to identify the connection of each device to the Internet
- **32-bit address** that uniquely and universally defines the connection of a host or a router to the Internet.
- IP address is the **address of the connection**, not the host or the router, because if the device is moved to another network, the IP address may be changed.
- If a **device has two connections** to the Internet, via two networks, it has **two IPv4 addresses**
- **Address Space:** the total number of addresses used by the protocol; the address space is 2^{32} or 4,294,967,296; **more than 4 billion devices** could be connected to the Internet.

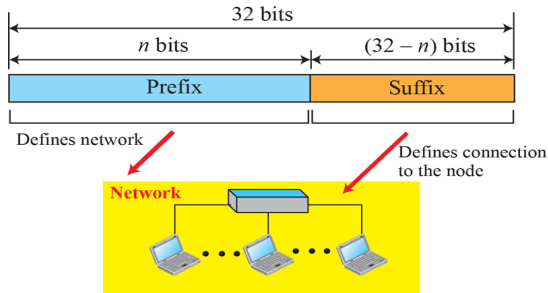
Notation

- Three common notations to show an IPv4 address: binary notation (base 2), dotted-decimal notation (base 256), and hexadecimal notation (base 16).



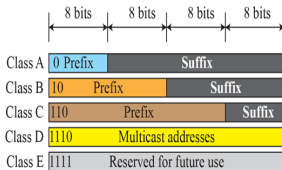
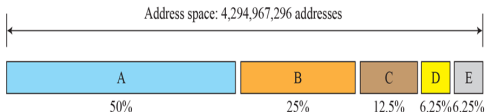
Hierarchy in Addressing

- 32-bit IPv4 address is hierarchical, divided only into two parts.
- The first part called the **prefix**, **defines the network**; the second part called the **suffix**, **defines the node** (connection of a device to the Internet).
- a fixed-length prefix: **classful addressing**.
- a variable-length network prefix : referred to as **classless addressing**



Classful Addressing

- Three fixed-length prefixes were designed instead of one ($n = 8$, $n = 16$, and $n = 24$).
- The whole address space was divided into five classes (class A, B, C, D, and E),



Class	Prefixes	First byte
A	$n = 8$ bits	0 to 127
B	$n = 16$ bits	128 to 191
C	$n = 24$ bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

Address Depletion

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

- The reason that **classful addressing** has become **obsolete** is **address depletion**.
- Addresses were not distributed properly; the addresses being rapidly used up
- class A can be assigned to only 128 organizations in the world, but each organization needs to have a single network with 16,777,216 nodes
- Class B addresses were designed for midsize organizations
- Class C :number of addresses that can be used in each network (256) was so small
- Class E addresses were almost never used, wasting the whole class.

Subnetting and Supernetting

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

- A class A or class B block is divided into several subnets.
- Each subnet has a larger prefix length than the original network.
- if a network in class A is divided into four subnets, each subnet has a prefix of $n_{sub} = 10$.
- If all of the addresses in a network are not used, subnetting divides the addresses among several organizations.
- While subnetting was devised to divide a **large block into smaller ones**, supernetting was devised to **combine several class C blocks into a larger block** to be attractive to organizations that need more than the 256 addresses available in a class C block

Advantage of Classful Addressing

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

- Classful addressing had several problems and became obsolete
- Given an address, we can easily find the class of the address and,
- since the prefix length for each class is fixed, we can find the prefix length immediately.
- Prefix length in classful addressing is inherent in the address; no extra information is needed to extract the prefix and the suffix.

Classless Addressing

Packet
Switching

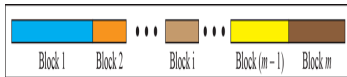
Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPv4
ADDRESSES

- larger address : a long-term solution — increase the length of IP addresses : IPv6
- Short-term solution : classless addressing — the class privilege was removed from the distribution to compensate for the address depletion.
- In classless addressing, variable-length blocks are used that belong to no classes
- Can have a block of 1 address, 2 addresses, 4 addresses, 128 addresses, and so on
- the whole address space is divided into variable length blocks.
- The prefix in an address defines the block (network); the suffix defines the node (device).
- Restriction: number of addresses in a block needs to be a power of 2.

- The prefix length in classless addressing is variable.
- A prefix length that ranges from 0 to 32.
- The size of the network is inversely proportional to the length of the prefix.
- A small prefix means a larger network; a large prefix means a smaller network.



Address space

Prefix Length: Slash Notation

- How to find the prefix length if an address is given.
- We need to separately give the length of the prefix, separated by a slash: slash notation and formally as **classless interdomain routing or CIDR** strategy.



Prefix
length

An arrow points from the text "Prefix length" to the yellow box containing "n".

Examples:

12.24.76.8/8

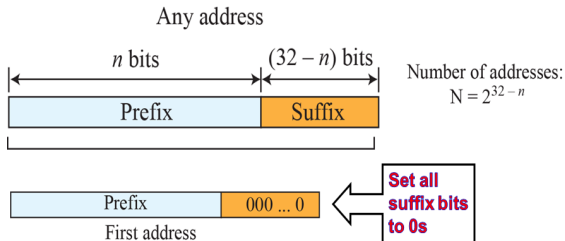
23.14.67.92/12

220.8.24.255/25

Extracting Information from an Address

To know three pieces of information about the **block to which the address belongs: the number of addresses, the first address in the block, and the last address.**

- 1 The number of addresses in the block is found as $N = 2^{32-n}$.
- 2 To find the first address, we keep the n leftmost bits and set the $(32 - n)$ rightmost bits all to 0s.
- 3 To find the last address, we keep the n leftmost bits and set the $(32 - n)$ rightmost bits all to 1s.



A classless address is given as 167.199.170.82/27. We can find the above three pieces of information as follows. The number of addresses in the network is $2^{32-n} = 25 = 32$ addresses. The first address can be found by keeping the first 27 bits and changing the rest of the bits to 0s.

Address: 167.199.170.82/27	10100111	11000111	10101010	01010010
First address: 167.199.170.64/27	10100111	11000111	10101010	01000000

Address: 167.199.170.82/27	10100111	11000111	10101010	01011111
Last address: 167.199.170.95/27	10100111	11000111	10101010	01011111

Address Mask

- Address mask is a 32-bit number in which the **n** leftmost bits are set to **1s** and the rest of the bits (**32 - n**) are set to **0s**.
- The A computer can easily find the address mask because it is the complement of $(2^{32-n} - 1)$.
- 1 The number of addresses in the block $N = \text{NOT}(\text{mask}) + 1$.
- 2 The first address in the block = (Any address in the block) AND (mask).
- 3 The last address in the block = (Any address in the block) OR [(NOT (mask))].

Example

Packet Switching

Unit-III

NETWORK-LAYER PERFORMANCE

IPV4 ADDRESSES

We repeat previous example using the mask. The mask in dotted-decimal notation is 256.256.256.224

..

Number of addresses in the block: $N = \text{NOT}(\text{mask}) + 1 = 0.0.0.31 + 1 = 32$ addresses

First address: $\text{First} = (\text{address}) \text{ AND } (\text{mask}) = 167.199.170.82$

Last address: $\text{Last} = (\text{address}) \text{ OR } (\text{NOT mask}) = 167.199.170.255$

Example

In classless addressing, an address cannot per se define the block the address belongs to. For example, the address 230.8.24.56 can belong to many blocks. Some of them are shown below with the value of the prefix associated with that block.

Prefix length:16	→	Block:	230.8.0.0	to	230.8.255.255
Prefix length:20	→	Block:	230.8.16.0	to	230.8.31.255
Prefix length:26	→	Block:	230.8.24.0	to	230.8.24.63
Prefix length:27	→	Block:	230.8.24.32	to	230.8.24.63
Prefix length:29	→	Block:	230.8.24.56	to	230.8.24.63
Prefix length:31	→	Block:	230.8.24.56	to	230.8.24.57

Network Address

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

- The **first address, the network address**, is particularly important because it is used in routing a packet to its destination network.
- When a packet arrives at the router from any source host, the **router needs to know to which network the packet should be sent**: from which interface the packet should be sent out.
- After the network address has been found, the router consults its forwarding table to find the corresponding interface from which the packet should be sent out.
- The **network address is actually the identifier of the network**

Block Allocation

Packet
Switching

Unit-III

NETWORK-
LAYER
PERFOR-
MANCE

IPV4
ADDRESSES

How are the blocks allocated?

- Block allocation is given to a global authority— Internet Corporation for Assigned Names and Numbers (ICANN)
- It assigns a large block of addresses to an ISP
- Two restrictions need to be applied to the allocated block
-